Neal Sharma

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EDUCATION

Iowa State University **Ames, Iowa**

Computer Science Honors - Minors in AI, Cybersecurity, and Data Science **Expected: May 2028**

GPA: 3.84/4.0 - Dean's List (All Semesters) **Junior**

EXPERIENCE

Application Development Intern — Summer 2026

Fareway Stores

- Developed a full-stack application used across 144 Fareway stores by leadership to track and analyze thousands of responses.
- Led project planning translating business requirements through discovery, sprint planning, user stories, and business demos.
- Built the Planogram Tracker (visual layout diagrams for shelf product placement) using Laravel/PHP, React, PostgreSQL, Redis, and external APIs, designing database structures, frontend components, backend, sync workflows, and role-based access.
- Collaborated with the web team through bug fixes, bash scripting, containerization with Docker, and production maintenance.

Software Engineer Intern — Summer 2025

Tata Consultancy Services

- Developed a full-stack retail management application using C#, .NET MVC, SQL Server, and jQuery (GitHub).
- Implemented authentication, image uploads, encryption, centralized logging, and MVC architecture linked to SQL Server.
- Rotated with frontend, backend, database, DevOps, and cloud engineers, gaining exposure to the full software development lifecycle and Agile workflows while learning skills in API development, Bootstrap, documentation, and Azure cloud.

Cybersecurity & Software Development Assistant — January 2026 - Present

ISEAGE - Cybersecurity Innovation and Outreach, Iowa State University

- Plan and execute multiple Cyber Defense Challenges with 100+ participants per competition by developing multi-application scenarios that simulate realistic operations, including web applications, services, and infrastructure.
- Gain hands-on experience across diverse programming languages, frameworks, Linux administration, networking, and cybersecurity by building and connecting services across different Docker containers and virtual machines.
- Lead live competitions alongside cybersecurity professionals, coordinating the event and managing infrastructure.

Mission Control - Operations Team — August 2026 - Present

Affordable Rocket Engineering Services (ARES) - Make to Innovate, Iowa State University

- Built a Linux-hosted Go dashboard for live rocket launches, providing real-time visualization of telemetry including altitude, temperature, velocity, and other flight data through WebSocket streaming.
- Integrated live GPS tracking and map visualization to display the rocket's exact location during flight, supporting improved mission monitoring and recovery operations.
- Designed the dashboard to support four distinct rocket platforms, enabling operators to switch between rocket configurations.

Research - Unlearning Attacks against Predictive Uncertainty in AI Models — August 2025 - May 2026

- Researched machine unlearning impacts on AI models - presented at the *National Conference on Undergraduate Research*.
- Implemented and compared methods for data removal from pre-trained image classification models.
- Applied uncertainty estimation techniques (Deep Ensembles, Monte Carlo Dropout) and metrics such as Expected Calibration Error and Brier Score to predict model behavior, confidence, accuracy, and identify vulnerabilities introduced by unlearning.

PROJECTS [nealks.netlify.app/portfolio/projects]

Hospital Management Platform [Android GitHub] [WebApp GitHub]

- Android: Awarded *Best Project Overall (1st place) among 300+ students*. Developed the backend using Spring Boot to implement appointments, prescriptions, pharmacy, Firebase notifications, video calling, and a Gemini-based AI health assistant.
- Web: Developed both frontend and backend for the React/Node.js application, building patient dashboards, doctor portals, pharmacy workflows, admin management, JWT authentication, MongoDB APIs, notifications, and health record features.

Home Server Infrastructure [GitHub]

- Built a local LLM setup with llama.cpp, running open-weight models with server read/write and internet access.
- Developed a centralized webapp with React and .NET to manage AI servers and storage, using Docker and GitHub deployments.
- Built cloud-like storage with multi-drive backups, SMB shares, file versioning, remote access, and a 3D-printed enclosure.

Space-Miner - 2D Java Game [GitHub]

- A top-down survival/adventure game built entirely from scratch in Java, under active development with plans for public release.
- Developed modular gameplay and engine systems including collision, procedural generation, culling, lighting, NPCs, UI, minimap, save/load, audio, and debugging, improving scalability and responsiveness.

SKILLS

Languages: Java, Python, C/C++, C#, PHP, TypeScript, JavaScript, SQL, Bash, Go, LEGv8 Assembly, HTML/CSS, LaTeX

Tools and Frameworks: Spring Boot, .NET, Laravel, Django, React, Node.js, Docker, Redis, Azure, Postman